

CDA 3103 Computer Organization Homework

Section I: Problems

1.

0x004F05BC	“h”
0x004F05BD	“e”
0x004F05BE	“l”
0x004F05BF	“l”
0x004F05C0	“o”
0x004F05C1	“ ”
0x004F05C2	“R”
0x004F05C3	“I”
0x004F05C4	“S”
0x004F05C5	“C”
0x004F05C6	“-“
0x004F05C7	“V”
0x004F05C8	“\0”

2. **or** s3, s4, s5

xori s3, s3, 1

3. **and** s3, s4, s5

xori s3, s3, 1

4.

a. **add** s7, s8, s9

Machine Code:

$s7 = s8 + s9$

R-type Instruction:

funct7	rs2	rs1	funct3	rd	opcode
7 bits	5 bits	5 bits	3 bits	5 bits	7 bits

Binary Code:

0000000	11001	11000	000	10111	0110011
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Hexadecimal Code: 0x019C0BB3

b. addi s3, s4, 28

Machine Code:

$s3 = s4 + 28$

I-type Instruction:

imm[11:0]	rs1	funct3	rd	opcode
12 bits.	5 bits	3 bits	5 bits	7 bits

Binary Code:

000000011100	10100	000	10011	0010011
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Hexadecimal Code: 0x01CA0993

c. sw s9, 16(t4)

Machine Code:

Memory $[t4 + 16] = s9$

S-Type Instruction:

imm[11:5]	rs2	rs1	funct3	imm[0:4]	opcode
7 bits	5 bits	5 bits	3 bits	5 bits	7 bits

Binary Code:

0000000	11001	11101	010	10000	0100011
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Hexadecimal Code: 0x019EA823

d. lw s4, 0x5C(t3)

Machine Code:

s4 = Memory [t3 + 92]

I-Type Instruction:

imm [11:0]	rs1	funct3	rd	opcode
12 bits	5 bits	3 bits	5 bits	7 bits

Binary Code:

000001011100	11100	010	10100	0000011
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Hexadecimal Code: 0x05CE2A03

e. srai t0, t1, 0xC

Machine Code:

t0 = t1 / 4096

I-Type Instruction:

imm [11:0]	rs1	funct3	rd	opcode
12 bits	5 bits	3 bits	5 bits	7 bits

Binary Code:

0100000	01100	00110	101	00101	0010011
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Hexadecimal Code: 0x40C35293

5.

a. C – Code

```
if (g > h)
    g = g + 1;
else
```

$h = h - 1;$

RISC - V

IF: bge a1, a0, ELSE

**THEN: addi, a0, a0, 1
j ENDIF**

ELSE: addi, a1, a1, -1

ENDIF:

b. C – Code

```
if (g <= h)
    g = g + h;
else
    g = g - h;
```

RISC – V

IF: blt a0, a1, ELSE

**THEN: add a0, a0, a1
j ENDIF**

ELSE: sub a0, a0, a1

ENDIF:

6. a. 29

addi s7, zero, 29

b. -214

addi s7, zero, -214

c. 0xEEEEEFAB

lui s7, 0xEEEEF

addi s7, s7, -85

d. 0xEDCBA123

lui s7, 0xEDCBA

addi s7, s7, 0x123

7. C - Code

```
int i;  
int temp[10];  
for ( i = 0; i < 10; i = i + 1)  
    temp[i] = i * 128;
```

RISC - V

addi x7, x0, 0 // i = 0

addi x8, x0, 10 // size = 10

FOR: bge x7, x8, EXIT

sllx20, x7, 7

sllx9, x7, 2

add x9, a0, x9

sw x20, 0(x9)

addi x7, x7, 1

j FOR

EXIT: